

# NIKUL PATEL

[pnikul165@gmail.com](mailto:pnikul165@gmail.com) | [LinkedIn](#) [n1kul] | [+1\(315\) 603-7990](tel:+13156037990) | Open to Relocate



## EDUCATION

### Clarkson University

Double Major: Aerospace and Mechanical Engineering | Minor: Robotics | Honors Program

Clubs and Organizations: CUSA | SASE | Underwater Robotics

Potsdam, NY

*Expected Graduation: 2028*

GPA: 3.98

## PROFESSIONAL EXPERIENCE

### Co-Op Student

Jan 2026 – July 2026

Baldwin Richardson Foods

Rochester, NY

- Supported Lean Six Sigma continuous improvement initiatives by participating in 2 Kaizen events, conducting 10+ root cause analyses, and implementing corrective actions to improve operational efficiency, reliability, and standardized work practices.
- Led a DMAIC-based LOTO standardization project by identifying process gaps, updating 50+ work instructions and compliance documents, improving operator training, and establishing sustainable processes to support long-term compliance.

### Undergraduate Researcher

Dec 2025 – Present

Space Dynamics Simulation

Potsdam, NY

- Currently in early-stage design and system definition of a 3DOF air-bearing spacecraft simulator to enable thruster and attitude control testing using hardware in-loop simulation
- Optimizing mass distribution and center of mass in SolidWorks and defining integrated electrical and pneumatic layouts (thrusters, air systems, control hardware) to meet stability and performance requirements

### Design Build Captain

Aug 2025 – Present

Underwater Robotics Team

Potsdam, NY

- Designing and 3D-printing custom fixtures and mechanical supports using SolidWorks and OnShape to improve subsystem integration and assembly reliability
- Collaborating with electrical and controls teams during subsystem integration, supporting wiring layouts, dimensional fit verification, assembly activities, and DFMA improvements

### Research Intern

Mar 2024 – July 2024

Spartificial

Remote

- Analyzed and organized a dataset of 22,000+ samples across 22 classes and 4 timeframes; independently designed a Late Fusion multi-view CNN while leading a 2-member team on Early & Intermediate Fusion models
- Achieved 95%+ accuracy for each model and 20% faster training, while optimization experiments improved predictive accuracy by 10% and processing efficiency by 15%

## LEADERSHIP AND ACADEMIC ENGAGEMENT

\*Most Recently held Position

**Teaching Assistant** | Honors MATLAB (HP102/103) | Clarkson University

Aug 2025 – Present

- Co-Managing a fully TA-led honors course | Coordinating lectures, curriculum, assignments, grading, and office hours

**Class Senator** | Student Association | Clarkson University

Sept 2025 – Present

- Advocating student interests through active participation in policy discussions and voting on key campus issues

**President\*** | SASE (Society of Asian Scientists and Engineers) | Clarkson Chapter

Sept 2024 – Present

- Leading Clarkson SASE chapter operations, collaborating with executive board members to advance organizational goals

**Student Manager\*** | Undergraduate Admissions | Clarkson University

Sept 2024 – Present

- Coordinated outreach operations, student engagement initiatives, and communication efforts for prospective students

## SKILLS

\*Certified Skill

**Design & Prototyping:** SolidWorks\*, OnShape, GD&T\*, DFMA\*, Technical Drawings & Documentation, Rapid Prototyping, FDM/3D Printing\*, Custom Fixture Design

**Analysis & Simulation:** Ansys\*, FEA, Thermal Analysis, Gazebo, Hardware-in-the-Loop (HIL) Simulation

**Robotics & Hardware:** Mechatronics & Electromechanical Systems, Arduino Embedded Systems, Pneumatic Systems, Sensor Integration, Actuators & Motors

**Programming & Software:** Python\*, MATLAB/Simulink\*, Deep Learning & CNNs (PyTorch/TensorFlow), PLC exposure.

**Continuous Improvement & Operations:** Lean Six Sigma\*, DMAIC, Kaizen, RCA, KPI Analysis, SOP Development

## AWARDS

**Presidential Scholar:** Academic Excellence | Clarkson University

All Semesters

**SASE Rising Leader:** Recognition for strong leadership skills and proactive initiative | Clarkson University

April 2025